

PROJECT EVOLUTION LOG

QuietMel Project Iteration History

Complete development history from the initial MVP to cloud-served help manuals

A code-backed operating baseline for future maintainers

Recorded period: 3-10 August 2026

Code baseline: main / 0ffe2b3

Evidence: Git history, current files, migrations, model outputs, and deployment configuration

Each entry distinguishes what was introduced from what remains current

How to read this history

This log expands the Git commit subjects into product goals, code changes, replacements, and current status. Merge commits and deployment-marker commits are retained for traceability, but feature truth comes from the final code and the non-merge commits.

Current endpoint

As of 10 August 2026, QuietMel includes live pedestrian data, persistent sensor locations, a database-backed six-hour forecast, multi-route quietness ranking, nearby quiet-place discovery, location-aware navigation, settings, onboarding, and private Supabase help manuals. Authentication has been deliberately removed.

1. Iteration stages

Stage	Date	Outcome	Current status
MVP shell	03 Aug	Responsive map, static crowd and route prototypes	Map shell remains; static crowd data replaced
Backend and live data	05 Aug	Express, Supabase, live sensors, three map layers	Core remains; authentication removed later
Place scenes	06 Aug	Google place categories and preferences	Categories remain; preferences moved to localStorage
Forecast implementation	08 Aug	Two-year model, database profiles, six-hour animation	Current runtime baseline
Route planning	08 Aug	Google alternatives plus quiet-route scoring	Expanded to selectable multi-route UI
Product completion	09 Aug	Settings, tour, nearby quiet places, deployment fixes	Current interaction baseline
Help centre	10 Aug	Static PDFs followed by private cloud delivery	Runtime is backend-only

2. Complete commit-by-commit record

Commit	Date	Subject	Implemented change	Current relationship
69f5be6	03 Aug	Initialize QuietMel MVP	Created the React/Vite/TypeScript project, responsive map, static route/place/crowd prototypes, styles, README, and product brief.	Static crowd sources were replaced; some demo assets remain.
b905143	05 Aug	Add authentication and live pedestrian sensor	Added Express, Supabase clients, health RPC, pedestrian_sensors migration, City live API, sensor	Live architecture remains; auth and Turnstile were later

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Commit	Date	Subject	Implemented change	Current relationship
		map	catalogue, refresh, layers, AuthPanel, Turnstile, and auth routes.	removed.
4b4e2e7	05 Aug	Change dev run on network automatically	Updated development scripts to expose Vite on the network and run frontend and backend together through concurrently.	Still used by npm run dev.
03babd0	06 Aug	Quick scene areas and route planning	Added eleven Google Place categories, quick filters, place walking routes, a user preference API/table, and forecast training documentation.	Place scenes remain; server-side user preferences were removed.
ce276d8	06 Aug	Delete tmp directory	Removed document-generation scripts, rendered PNGs, intermediate PDFs, and temporary DOCX files.	Repository cleanup.
9f575ba	07 Aug	Fix duplicate mobile filter button	Corrected overlapping desktop/mobile category controls and removed remaining Office lock and temporary files.	Mobile fix remains.
9cd5ffe	07 Aug	Merge main	Synchronised branch history.	Merge node only.
37e366f	08 Aug	Remove login and implement six-hour forecast	Deleted auth, Turnstile, backend preferences, and the user preference migration. Added the training pipeline, 67,200 profiles, validation metrics, forecast migration/import, API, and 25-frame UI.	Defines the no-account product and current forecast baseline.
865dbda	08 Aug	Merge pull request #2	Merged the no-login and forecast implementation branch.	Merge node.
0b2582b	08 Aug	First navigation version	Added origin/destination search, Google route alternatives, Express crowd scoring, route sheet, start/end navigation, and real route overlays.	Later refined for multi-route selection and UI.
8697875	09 Aug	Landing instructions and project completion	Completed route ranking and solid/dashed rendering; added settings, Original/Dark themes, language entry, five-step tour, nearby quiet place, permission gating, and UI refinements.	Current product UX baseline; localisation remains partial.
5c7ebe7	09 Aug	Merge pull request #4	Merged navigation and product-completion work.	Merge node.
f6147c1	09 Aug	Exclude local model assets	Added .vercelignore so model training assets are not uploaded to Vercel.	Later expanded to exclude output/.
f192cff	09 Aug	Prepare complete QuietMel release	Recorded a production-release preparation milestone with no separate file diff.	Deployment marker.
e7cab7	09 Aug	Route Vercel API requests to Express	First attempted a catch-all api/[...path].ts Function for production API routing.	Replaced in the next commit.
ae3a527	09 Aug	Route production API requests to Express	Removed the catch-all Function and added vercel.json rewrites from /api/:path* to the single Express entry.	Current production routing design.
f2955e2	10 Aug	Add user manuals	Generated three two-page PDFs and added /help, a manual sidebar, downloads, settings entry, and Safari autocomplete click protection.	Help UI remains; public static PDFs were removed next.
0ffe2b3	10 Aug	Serve manuals from the	Added help_manuals, a private bucket, help API,	Current HEAD and help-

Commit	Date	Subject	Implemented change	Current relationship
		cloud	one-hour signed URLs, upload script, and removed all local runtime PDF fallback.	delivery design.

3. Feature lifecycle

3.1 Crowd data: prototype to live to forecast

1. 03 Aug: sensoryPressure.ts and sensoryForecast.ts provided static demo heat data.
2. 05 Aug: City of Melbourne past-hour minute counts replaced static live data; a 60-second cache and stable sensor table were added.
3. 08 Aug: hourly-gradient-v1 added a two-year historical model. Runtime prediction was changed to read Supabase only, with no local model fallback.
4. Current: live heatmap is default; forecast is requested on demand and animates six future hours; zero values draw no heat.

3.2 Identity and preferences: added, simplified, removed

5. 05 Aug: email authentication, Turnstile, AuthPanel, and auth routes were introduced.
6. 06 Aug: user_map_preferences and a preference API were added.
7. 08 Aug: the product moved to a no-login model; auth, CAPTCHA, preference table, and backend preference routes were removed.
8. Current: theme, language, tour completion, and quick categories are stored only in localStorage.

3.3 Routes: demo to Google alternatives to crowd ranking

9. The MVP used demoRoutes for route visualisation.
10. The first real route version requested Google walking alternatives and sent compact geometry to Express for live-plus-forecast scoring.
11. The completed version exposes multiple ranked choices, a solid selected line, dashed alternatives, priority, overall score, average/peak crowd, and coverage.
12. Nearby quiet-place routing is a separate fast path: semantic place category plus distance, then Google's fastest route without quiet-route scoring.

3.4 Help delivery: static public files to private cloud

13. The first 10 Aug version placed PDF copies in both output/pdf and public/help.
14. The second version deleted public/help and added Supabase metadata and a private Storage bucket.

15. HelpLibrary now loads a catalogue and then requests a one-hour signed URL for the selected manual. There is no local runtime fallback.
16. output/pdf remains an upload source only and is excluded from Vercel.

4. Database migration history

Migration	Object	Purpose	Current status
20260805000100_health_check.sql	public.health_check()	Verify Express-to-Supabase connectivity	Retained; service_role execute only
20260805000200_pedestrian_sensors.sql	pedestrian_sensors	Stable sensor catalogue and source fallback	Retained with RLS
20260806000100_user_map_preferences.sql	User preference table	Cross-device preferences during login phase	Deleted from repository; should not be used
20260808000100_crowd_hourly_profiles.sql	crowd_hourly_profiles	Six-hour forecast profiles	Retained; removes 28, 65, 78
20260810000100_help_manuals.sql	help_manuals and private bucket	Cloud help centre	Retained; 10 MB PDF-only bucket

Migration-history warning

Deleting a migration file does not remove an object that was already created in an early Supabase project. If user_map_preferences still exists online, remove it only through a new explicit drop migration after confirming that no data or code depends on it.

5. Model and data artifacts

File	Date / range	Purpose	Deployment status
model/train_crowd_profiles.py	07 Aug 2026	Download, clean, grid, fallback, gradient, validate	Not deployed
model/output/crowd_hourly_profiles.csv	07 Aug 2024 - 06 Aug 2026	67,200-row import contract	Imported to Supabase; not deployed
model/output/model_metadata.json	07 Aug 2026	Window, quality, row counts, version audit	Not deployed
model/output/validation_metrics.cs	Latest 28-day holdout	MAE/WAPE for 99 test sensors	Not deployed

File	Date / range	Purpose	Deployment status
v			
scripts/import-forecast.mjs	08 Aug 2026	Validate and batch-import profiles	Maintenance command
output/pdf/*.pdf	10 Aug 2026	Manual upload source	Not deployed; Storage is runtime source
scripts/upload-help-manuals.mjs	10 Aug 2026	Upload PDFs and upsert metadata	Maintenance command

6. UI and interaction evolution

Feature	Final behaviour	Primary files
Live refresh	Fixed map control, loading animation, refresh=1 no-cache request	CrowdRefreshButton.tsx; useLiveCrowd.ts
Layer control	Heatmap/Sensors/Combined dialog with sensor availability signal	MapLayersControl.tsx
Sensor details	Marker click, close button, 220 ms pointer-leave close, dynamic Google summary	MapView.tsx
Search panel	Collapsed original search; expanded start/destination; remains while loading; closes on route ready	RouteSearchPanel.tsx
Safari autocomplete	pointerdown prevents input blur from removing suggestions before click	RouteSearchPanel.tsx
Multi-route map	Recommended solid line, alternatives dashed, map/list selection	MapView.tsx; RoutePlanner.tsx
Nearby quiet place	Fixed bottom-right control, location gate, dedicated loader	App.tsx; app.css
Settings	Top-right dialog with Original/Dark, language, tutorial, and help	AppSettings.tsx
First-run tour	Modal shade blocks background and tracks five target bounds	OnboardingTour.tsx
Help reader	New page, document sidebar, private signed PDF object	HelpLibrary.tsx; help.css

7. Deployment and production-routing fixes

- The first Vercel release served the frontend but returned 404 for `/api/crowd/live` and `/api/crowd/forecast`.

- A catch-all Function was attempted first; the final design uses one `api/index.ts` Express entry plus `vercel.json` rewrites.
- Current rewrite order is `/api/:path* -> /api`, followed by `/:path* -> /index.html`.
- `model/` and `output/` are excluded to reduce the deployment bundle and enforce database-backed runtime data.
- The 10 Aug production project is `quietmel`. The recorded deployment URL is `quietmel-9smhc4yco-nariyirms-projects.vercel.app` and the stable domain is `quietmel.vercel.app`.

8. Replaced or removed files

Removed / superseded item	Original purpose	Replacement
<code>src/data/sensoryPressure.ts</code>	Simulated live heatmap	City of Melbourne live API
<code>sensoryForecast.ts</code> / <code>SensoryForecast.tsx</code>	Simulated forecast	Supabase hourly-gradient-v1 and <code>ForecastControls</code>
<code>server/routes/auth.ts</code> / <code>AuthPanel</code> / <code>auth.ts</code>	Email authentication	No-login product
<code>TurnstileWidget.tsx</code>	Cloudflare human verification	Removed with authentication
<code>server/routes/preferences.ts</code>	Cloud user category preferences	<code>quietmel.map-preferences</code> <code>localStorage</code>
<code>user_map_preferences</code> migration	User preference table	No longer required; explicit drop if still online
<code>api/[...path].ts</code>	Vercel API catch-all	<code>vercel.json</code> rewrite plus <code>api/index.ts</code>
<code>public/help/*.pdf</code>	Static help delivery	Private Supabase Storage and signed URLs
<code>tmp/forecast-guide-render/*</code>	Document QA intermediates	Deleted; not runtime assets

9. Current repository inventory

- Runtime code: `src/`, `server/`, and `api/`.
- Database: four current SQL files under `supabase/migrations/`.
- Model: `model/README.md`, `train_crowd_profiles.py`, `requirements.txt`, and three output artifacts.
- Maintenance scripts: `import-forecast.mjs`, `upload-help-manuals.mjs`, and `generate-help-manuals.py`.
- Documentation: `README.md`, `PRODUCT.md`, forecast training documentation, team information, and the English handover documents.
- Deployment: `vercel.json`, `.vercelignore`, and a local non-committed `.vercel` project link.

10. Template for future iteration entries

Field	Required record
Version / date / commit	A unique pointer to the code
Product objective	The user problem and scope
Code changes	Primary files and modules
Data contract	Added/removed fields, units, enums, thresholds
Database migration	Filename, production status, rollback
Third-party configuration	Google, Supabase, or Vercel changes
Test evidence	Lint, build, API, browser, and device checks
Deployment	Production URL, deployment ID, environment changes
Compatibility and risk	Known follow-up work

Change-log rule

Any change that affects displayed metrics, forecast outputs, database structure, API responses, privacy boundaries, or production routing must also update the maintenance handbook.

End of iteration history.